

MAT21-1

Linear Algebra II

Albert School — 34 hours

Preface

In **Linear Algebra I** (MAT11-3) you learned to compute with matrices and to read them as **linear maps**. You solved $AX = B$ by Gaussian elimination, you found the kernel and the column space of a matrix, and — crucially for this course — you saw that one linear map can be written as **different matrices** depending on the basis you choose. The change-of-basis machinery was introduced there with a promise: that picking the right basis can make a map dramatically simpler. This course delivers on that promise.

The guiding question is this. A square matrix A describes a transformation of space, but the **numbers** in A depend on an arbitrary choice of coordinates; they mix together what the map genuinely does with the accident of how we labelled the axes. Is there a way to see past the coordinates to the transformation itself? The answer is yes, and it rests on a single idea: almost every transformation has **special directions it does not rotate** — it merely stretches or shrinks vectors lying along them. These are the **eigenvectors**, and the stretch factors are the **eigenvalues**. Choose your axes **along** these directions and the matrix collapses to a diagonal one, a list of independent scalings. That is **diagonalization**, and it turns hard questions — what is A^{100} ? how does a population, a random walk, or a recurrence behave in the long run? — into easy ones.

We build the tools in order. First the **determinant**, a single number measuring how a map scales volume and, as a by-product, detecting exactly when a map is invertible — the gate through which eigenvalues are found. Then **eigenvalues and eigenvectors** themselves, and the subtle counting (**algebraic vs. geometric multiplicity**) that decides whether the diagonalization dream comes true. We assemble these into the **diagonalization** procedure and its spectacular payoff, matrix powers. The **Cayley–Hamilton theorem** gives a second, algebraic route to inverses and powers that works even when diagonalization fails. We close with the applications the whole edifice was built for — counting walks in **graphs** and the long-run behaviour of **Markov chains** — and with the conceptual capstone: **similar matrices** are one transformation seen in different bases, and diagonalization is simply the best basis there is.

Keep one slogan in view: **to understand a matrix, find the directions it leaves alone.**

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1 Determinants

Before eigenvalues we need a detector for invertibility: a single number, computable from the entries of a square matrix, that is non-zero **exactly** when the matrix is invertible. That number is the **determinant**. It has a vivid geometric meaning — it measures how the map $X \mapsto AX$ scales volume — and a toolbox of properties that make even large determinants tractable.

1.1 A first example: area in the plane

Start with the most concrete case. Take the matrix

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix},$$

and watch what its map does to the **unit square** — the square with corners 0 , $e_1 = (1, 0)$, $e_2 = (0, 1)$, and $(1, 1)$. The two basis vectors land on the two columns of A , namely $(3, 1)$ and $(1, 2)$, and the square is carried to the **parallelogram** they span.

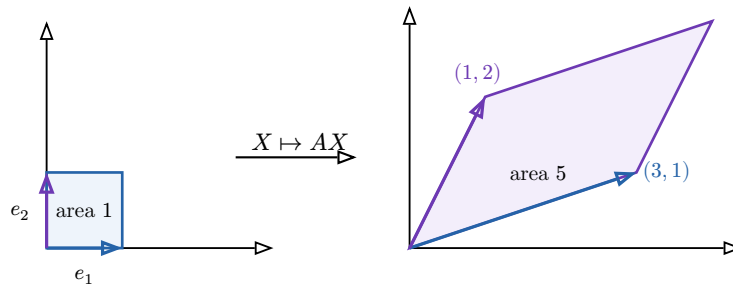


Figure 1: The map $X \mapsto AX$ sends the unit square (area 1) to the parallelogram spanned by the columns of $A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$. Its area is $3 \cdot 2 - 1 \cdot 1 = 5$ — exactly $\det A$. The determinant is the factor by which the map scales area.

The area of that parallelogram is $3 \cdot 2 - 1 \cdot 1 = 5$. This number — the **signed area** of the image of the unit square — is the determinant. Every other region’s area is scaled by the same factor 5, because the map is linear. We now make this precise and extend it to any size.

1.2 Computation

Definition 1 To each square matrix $A \in M_n(\mathbb{R})$ is attached a scalar $\det A$, its **determinant**. For $n = 1$, $\det(a) = a$. For $n = 2$,

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

For 3×3 matrices there is a convenient mnemonic, **Sarrus’ rule**: copy the first two columns to the right of the matrix, then add the three “down-right” diagonals and subtract the three “down-left” diagonals.

Proposition 2 (Sarrus’ rule)

$$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}.$$

Common pitfall Sarrus' rule works **only** for 3×3 matrices. There is no “Sarrus for 4×4 ” — the naive extension (six diagonal products) gives the wrong answer. For $n \geq 4$ use cofactor expansion or row reduction below.

For general n the determinant is built recursively, expanding along a row or column.

Definition 3 Let $A = (a_{ij}) \in M_n(\mathbb{R})$. The **minor** M_{ij} is the determinant of the $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j of A . The **cofactor** is $C_{ij} = (-1)^{i+j}M_{ij}$. Expansion along row i gives

$$\det A = \sum_{j=1}^n a_{ij}C_{ij},$$

and the analogous formula holds along any column j .

The signs $(-1)^{i+j}$ follow a checkerboard pattern starting with $+$ in the top-left corner:

$$\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}.$$

Worked example — cofactor expansion along the best row. Compute $\det A$ for

$$A = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 0 & 4 \\ 5 & 6 & 7 \end{pmatrix}.$$

Expanding along **column 2** is smartest — it has two zeros, killing two of the three terms. Only $a_{32} = 6$ survives, with sign $(-1)^{3+2} = -1$:

$$\det A = -6 \cdot \det \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} = -6(2 \cdot 4 - 1 \cdot 3) = -6 \cdot 5 = -30.$$

Common pitfall Always expand along the row or column with the **most zeros**. Each zero entry removes an entire cofactor from the sum, turning an n -term computation into a far shorter one. Expanding blindly along the first row when another row is mostly zeros is wasted effort — and extra chances to slip a sign.

The third method, **row reduction**, is the fastest for large matrices and rests on the properties below: reduce A to triangular form, tracking the effect of each operation, then read off the determinant as the product of the diagonal entries.

Proposition 4 (Determinant of a triangular matrix) If A is upper- or lower-triangular, $\det A$ is the product of its diagonal entries: $\det A = a_{11}a_{22}\dots a_{nn}$.

1.3 Geometric interpretation: signed volume

The opening example generalizes. In $\mathbb{R}^n$, the map $X \mapsto AX$ sends the unit n -cube to the parallelepiped spanned by the columns of A , and

$$\text{volume of image} = |\det A|.$$

The **sign** of $\det A$ carries extra information: it records whether the map **preserves** orientation ($\det A > 0$) or **reverses** it like a mirror ($\det A < 0$). A determinant of zero means the cube is flattened into something lower-dimensional — zero volume — which is the geometric face of non-invertibility.



Figure 2: The sign of the determinant is orientation. On the left the ordered pair (blue, purple) keeps its counter-clockwise sense — $\det > 0$. On the right the same two vectors appear in swapped angular order, a mirror image — $\det < 0$. A zero determinant is the boundary case where the two vectors line up and the area collapses to 0.

1.4 Properties

These identities are what make determinants usable; learn to reach for them before computing anything by brute force.

Proposition 5 (Algebraic properties) For $A, B \in M_n(\mathbb{R})$:

- **Multiplicativity:** $\det(AB) = \det(A) \det(B)$.
- **Transpose invariance:** $\det(A^T) = \det(A)$.
- **Invertibility criterion:** A is invertible $\iff \det A \neq 0$, and then $\det(A^{-1}) = 1/\det(A)$.
- $\det(I_n) = 1$ and $\det(\lambda A) = \lambda^n \det(A)$.

The multiplicativity rule is geometrically obvious once stated: applying B then A scales volume by $\det B$ then by $\det A$, so the combined factor is the product. Transpose invariance means every statement about **rows** below has an identical twin about **columns**.

Common pitfall Two seductive but **false** “rules”: in general $\det(A + B) \neq \det A + \det B$, and $\det(\lambda A) \neq \lambda \det A$ — the scalar comes out as λ^n , once per row. The determinant is multiplicative, never additive. Test on $A = B = I_2$: $\det(2I_2) = 4 \neq 2 = 2 \det(I_2)$.

Proposition 6 (Effect of row operations)

- **Swapping** two rows multiplies the determinant by -1 .
- **Scaling** a row by λ multiplies the determinant by λ .
- **Adding** a multiple of one row to another **leaves the determinant unchanged**.

Consequently $\det A = 0$ whenever two rows are equal, or a row is zero, or one row is a multiple of another.

The third operation is the workhorse: because it costs nothing, you may freely clear out entries to create zeros, then expand or triangularize.

Worked example — row reduction beats brute force.

$$\det \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 1 & 1 & 4 \end{pmatrix}.$$

Use $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$ (neither changes the determinant):

$$\det \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -1 & 1 \end{pmatrix}.$$

Now $R_3 \leftarrow R_3 + R_2$ (again free), giving an upper-triangular matrix:

$$\det \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{pmatrix} = 1 \cdot 1 \cdot 3 = 3.$$

Exercises

1. Compute $\det \begin{pmatrix} 4 & 3 \\ 6 & 5 \end{pmatrix}$ and $\det \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 1 \\ 2 & 1 & 0 \end{pmatrix}$.
2. Without expanding, explain why $\det \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 7 & 0 & 5 \end{pmatrix} = 0$.
3. If $A \in M_3(\mathbb{R})$ has $\det A = 4$, find $\det(2A)$, $\det(A^T)$, $\det(A^{-1})$, and $\det(A^3)$.
4. Use row reduction to compute the determinant of $\begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$, and confirm with Sarrus.

2 Eigenvalues and eigenvectors

We arrive at the central objects of the course. The determinant was built to detect invertibility; we now use it to hunt for the **special directions** a matrix does not rotate.

2.1 A first example: directions left alone

Consider

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

For a **typical** vector, A both rotates and stretches it. Take $x = (1, 0)$: then $Ax = (2, 1)$, which points in a different direction — the map has turned it. But try $v = (1, 1)$:

$$Av = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 3v.$$

The map did **not** rotate v at all; it merely stretched it by a factor of 3. The direction of $(1, 1)$ is **stable** under A . Another such direction is $w = (1, -1)$, for which $Aw = (1, -1) = 1 \cdot w$: here the factor is 1, so w is left completely fixed.

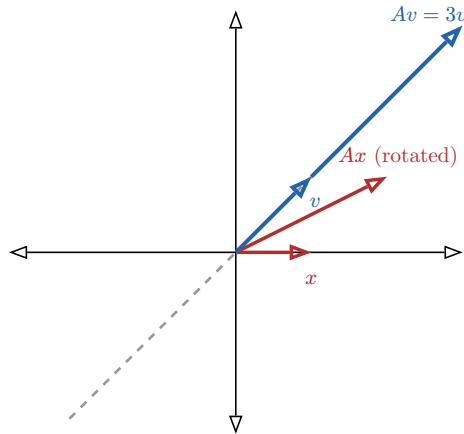


Figure 3: For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ a generic vector x is rotated off its line (red), but the eigenvector $v = (1, 1)$ stays on its line (dashed) and is only scaled, $Av = 3v$. The eigenvalue 3 is that scale factor; the line is the stable direction.

2.2 Definitions

Definition 7 Let $A \in M_n(\mathbb{R})$. A scalar $\lambda \in \mathbb{R}$ is an **eigenvalue** of A if there exists a **non-zero** vector v , called an **eigenvector**, with

$$Av = \lambda v.$$

The set

$$E_\lambda = \ker(A - \lambda I) = \{v \in \mathbb{R}^n : Av = \lambda v\}$$

is the **eigenspace** of λ . It is a subspace of $\mathbb{R}^n$, and its non-zero elements are exactly the eigenvectors for λ .

Common pitfall The vector v must be **non-zero** — otherwise $A0 = \lambda 0$ holds for every λ and the notion would be vacuous. By contrast the **eigenvalue** λ **may** be 0: 0 is an eigenvalue precisely when A has a non-trivial kernel, i.e. when A is **not** invertible. So “0 is an eigenvalue” and “ A is singular” say the same thing.

2.3 The characteristic polynomial

How do we **find** the eigenvalues? Rearrange $Av = \lambda v$ as $(A - \lambda I)v = 0$. This homogeneous system has a non-zero solution v exactly when $A - \lambda I$ is **not invertible** — and our invertibility detector is the determinant.

Definition 8 The **characteristic polynomial** of $A \in M_n(\mathbb{R})$ is

$$\chi_A(\lambda) = \det(A - \lambda I),$$

a polynomial of degree n in λ . The eigenvalues of A are exactly its **real roots**.

Worked example — the full computation. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$,

$$\chi_A(\lambda) = \det \begin{pmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{pmatrix} = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3).$$

The eigenvalues are $\lambda = 1$ and $\lambda = 3$, confirming the directions we found by hand. For $\lambda = 3$, solve $(A - 3I)v = 0$:

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 \implies v_1 = v_2,$$

so $E_3 = \text{span}((1, 1))$. Likewise $E_1 = \text{span}((1, -1))$.

Common pitfall Not every real matrix has real eigenvalues. The rotation by 90° , $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, has $\chi_A(\lambda) = \lambda^2 + 1$, with **no real roots** — geometrically obvious, since a 90° rotation leaves **no** direction unmoved. Over $\mathbb{R}$ such a matrix simply has no eigenvalues; this is exactly the situation the diagonalization criterion will flag.

2.4 Trace and determinant as sum and product

The characteristic polynomial ties the eigenvalues to two quantities you can read off A almost instantly, giving a free sanity check on every eigenvalue computation.

Proposition 9 Counting each eigenvalue with its algebraic multiplicity (as a root of χ_A), and writing $\lambda_1, \dots, \lambda_n$ for the n roots:

- the **trace** $\text{tr } A = a_{11} + \dots + a_{nn}$ equals $\lambda_1 + \dots + \lambda_n$;
- the **determinant** $\det A$ equals $\lambda_1 \dots \lambda_n$.

Worked example — checking eigenvalues with trace and determinant. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ we found eigenvalues 1 and 3. Check: $1 + 3 = 4 = \text{tr } A$ ✓ and $1 \cdot 3 = 3 = \det A$ ✓. Whenever you compute eigenvalues, verify their sum is the trace and their product the determinant — it catches most arithmetic slips on the spot.

Exercises

- Find the eigenvalues and a basis of each eigenspace for $A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$.
- Show that the eigenvalues of a triangular matrix are exactly its diagonal entries.
- A matrix $A \in M_2(\mathbb{R})$ has $\text{tr } A = 5$ and $\det A = 6$. Find its eigenvalues without knowing A .
- Explain why 0 is an eigenvalue of $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, and find E_0 .

3 Algebraic and geometric multiplicity

When eigenvalues repeat, a subtle counting question decides everything that follows. There are **two** ways to count how many times an eigenvalue “occurs”, and they need not agree.

3.1 Two ways to count, and why they differ

Definition 10 Let λ be an eigenvalue of A .

- Its **algebraic multiplicity** $m_a(\lambda)$ is its multiplicity as a root of the characteristic polynomial χ_A .
- Its **geometric multiplicity** $m_g(\lambda) = \dim E_\lambda$ is the dimension of its eigenspace — the number of independent eigenvectors it supplies.

The algebraic multiplicity counts the eigenvalue with its bookkeeping multiplicity in χ_A ; the geometric multiplicity counts the **actual independent directions** it gives you. The first is what the polynomial promises; the second is what the matrix delivers. They can differ.

Worked example — when the two disagree — the shear. Consider the **shear** $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Its characteristic polynomial is

$$\chi_A(\lambda) = \det \begin{pmatrix} 1-\lambda & 1 \\ 0 & 1-\lambda \end{pmatrix} = (1-\lambda)^2,$$

so $\lambda = 1$ has algebraic multiplicity $m_a = 2$. But the eigenspace is

$$E_1 = \ker \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \text{span}((1, 0)),$$

only **one** dimension: $m_g = 1$. The polynomial promised “two”, the matrix delivered “one”. This single eigenvector shortfall is what will make A **not diagonalizable**.

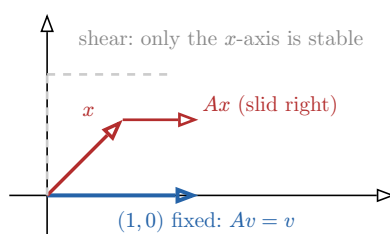


Figure 4: The shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ fixes every vector on the x -axis (blue) but slides everything else rightward in proportion to its height (red). Only one independent stable direction exists, so $m_g = 1$ even though $m_a = 2$. A shear is the prototype of a non-diagonalizable matrix.

3.2 The fundamental inequality

Proposition 11 For every eigenvalue λ of A ,

$$1 \leq m_g(\lambda) \leq m_a(\lambda).$$

The lower bound holds because an eigenvalue, by definition, has at least one eigenvector, so $\dim E_\lambda \geq 1$. The upper bound is the deeper fact: an eigenvalue can never supply **more** independent directions than its algebraic multiplicity allows. The interesting question is always whether the upper bound is **attained**.

Common pitfall The gap $m_g < m_a$ is the only obstruction to diagonalization, and it can only occur at a **repeated** eigenvalue. If $m_a(\lambda) = 1$ then the inequality forces $m_g(\lambda) = 1$ too, so simple eigenvalues are never a problem. Always scrutinize the **repeated** eigenvalues — that is where diagonalizability is won or lost.

4 Diagonalization

Now the payoff. If a matrix supplies **enough** eigenvectors to form a basis, then in that basis it becomes diagonal — a list of independent scalings — and almost every computation about it becomes trivial.

4.1 The idea: a basis of eigenvectors

Definition 12 A matrix $A \in M_n(\mathbb{R})$ is **diagonalizable** (over $\mathbb{R}$) if there exist an invertible matrix P and a diagonal matrix D with

$$A = PDP^{-1}.$$

The content of this equation is concrete. If $v_1, \dots, v_n$ are eigenvectors forming a basis, with $Av_i = \lambda_i v_i$, set $P = (v_1 \mid \dots \mid v_n)$ (eigenvectors as columns) and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ (eigenvalues on the diagonal, **in the same order**). Then $AP = PD$, which is $A = PDP^{-1}$. Reading it as a change of basis (Chapter 8): in the basis of eigenvectors, A is the diagonal matrix D — it just scales each axis independently.

Common pitfall The order must match: the i -th column of P must be an eigenvector for the i -th diagonal entry of D . Shuffling the eigenvalues without shuffling the eigenvectors the same way gives a wrong PDP^{-1} . Also, P 's columns must be **linearly independent** (so that P^{-1} exists) — which is exactly the diagonalizability question.

4.2 The criterion

Theorem 13 (Diagonalization criterion) $A \in M_n(\mathbb{R})$ is diagonalizable over $\mathbb{R}$ if and only if **both**:

1. the characteristic polynomial **splits** over $\mathbb{R}$ (factors into degree-one terms — equivalently, all n roots are real, counted with multiplicity); and
2. for **every** eigenvalue, $m_g(\lambda) = m_a(\lambda)$.

Condition 1 ensures there are n real eigenvalues to work with at all (recall the 90° rotation, which fails here). Condition 2 ensures the eigenvectors add up to a full basis: summing $m_g(\lambda) = m_a(\lambda)$ over all eigenvalues gives exactly n independent eigenvectors.

Proposition 14 (A convenient sufficient condition) If $A \in M_n(\mathbb{R})$ has n **distinct** real eigenvalues, then A is diagonalizable.

This is the common easy case: distinct eigenvalues each force $m_a = 1$, hence $m_g = 1$, and eigenvectors for distinct eigenvalues are automatically independent. You only need the full two-part criterion when eigenvalues repeat.

4.3 Procedure

To diagonalize A :

1. Compute $\chi_A(\lambda) = \det(A - \lambda I)$ and its real roots — the eigenvalues, with their algebraic multiplicities.
2. For each eigenvalue λ , solve $(A - \lambda I)v = 0$ to find a basis of E_λ ; its size is $m_g(\lambda)$.
3. If $\sum m_g(\lambda) = n$ (equivalently, the criterion holds), collect all these basis vectors as the columns of P and the matching eigenvalues into D . Then $A = PDP^{-1}$. Otherwise A is **not** diagonalizable.

Worked example — a complete diagonalization. Diagonalize $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. From Chapter 2, eigenvalues 3 and 1 with eigenvectors $(1, 1)$ and $(1, -1)$. Two distinct eigenvalues in $\mathbb{R}^2$, so A is diagonalizable. Take

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

Here $P^{-1} = -\frac{1}{2} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, and one checks $A = PDP^{-1}$.

Worked example — a matrix that is NOT diagonalizable. The shear $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has the single eigenvalue 1 with $m_a = 2$ but $m_g = 1$. Condition 2 fails, so A is **not** diagonalizable — there is no basis of eigenvectors, because there is only one independent eigenvector to be had.

4.4 Matrix powers — the headline application

Proposition 15 If $A = PDP^{-1}$ with $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, then for every integer $k \geq 0$,

$$A^k = PD^kP^{-1}, \quad \text{where} \quad D^k = \text{diag}(\lambda_1^k, \dots, \lambda_n^k).$$

Proof. The middle factors telescope: $A^k = (PDP^{-1})(PDP^{-1})\dots(PDP^{-1})$, and each interior $P^{-1}P = I$ cancels, leaving PD^kP^{-1} . A diagonal matrix is raised to a power entrywise. $\square$

This is the reason diagonalization matters in practice. Computing A^{100} directly is hopeless; via diagonalization it costs one matrix inversion and n scalar powers.

Worked example — a large power, instantly. With $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = PDP^{-1}$ as above,

$$A^k = P \begin{pmatrix} 3^k & 0 \\ 0 & 1 \end{pmatrix} P^{-1} = \frac{1}{2} \begin{pmatrix} 3^k + 1 & 3^k - 1 \\ 3^k - 1 & 3^k + 1 \end{pmatrix}.$$

No matter how large k is, this is a one-line formula.

4.5 Linear recurrence relations

A vector recurrence $u_{k+1} = Au_k$ has the immediate solution $u_k = A^k u_0$. When A is diagonalizable, $A^k = PD^kP^{-1}$ turns this into a **closed form** — each component of u_k becomes a combination of the powers λ_i^k .

Worked example — Fibonacci in closed form. The Fibonacci numbers $F_{k+1} = F_k + F_{k-1}$ become a vector recurrence by stacking two consecutive terms:

$$\begin{pmatrix} F_{k+1} \\ F_k \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} F_k \\ F_{k-1} \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Here $\chi_A(\lambda) = \lambda^2 - \lambda - 1$, with roots the golden ratio $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$ — distinct, so A is diagonalizable. Then $u_k = A^k u_0$ yields **Binet's formula**

$$F_k = \frac{\varphi^k - \psi^k}{\sqrt{5}}.$$

The growth rate φ is the **dominant eigenvalue**; the term ψ^k , with $|\psi| < 1$, decays away. The long-run behaviour is governed by the largest eigenvalue — a theme we return to in the applications.

Exercises

9. Diagonalize $A = \begin{pmatrix} 3 & 0 \\ 8 & -1 \end{pmatrix}$, or show it is not diagonalizable.
10. For $A = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$, find P, D and compute A^k in closed form.
11. Is $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$ diagonalizable? What are m_a and m_g of the eigenvalue 2? (Note A is already diagonal.)
12. Show that $\begin{pmatrix} 5 & 1 \\ 0 & 5 \end{pmatrix}$ is not diagonalizable.

5 The Cayley–Hamilton theorem

Diagonalization is powerful but not always available. The Cayley–Hamilton theorem gives a purely algebraic identity, valid for **every** square matrix, that yields inverses and powers even when no basis of eigenvectors exists.

5.1 Statement and verification

Theorem 16 (Cayley–Hamilton) Every square matrix satisfies its own characteristic polynomial:

$$\chi_A(A) = 0,$$

where the substitution replaces λ^k by A^k and the constant term c_0 by $c_0 I$.

Common pitfall Cayley–Hamilton does **not** say “ $\chi_A(A) = \det(A - AI) = \det(0) = 0$ ” — that reasoning is nonsense, because $\chi_A(A)$ substitutes the matrix A into the **polynomial** χ_A , it does not put A back inside the determinant. In particular each scalar coefficient c_k multiplies the matrix power A^k , and the constant term becomes $c_0 I$, never the scalar c_0 .

Worked example — verifying Cayley–Hamilton. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $\chi_A(\lambda) = \lambda^2 - 4\lambda + 3$. Substitute A :

$$A^2 - 4A + 3I = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} - \begin{pmatrix} 8 & 4 \\ 4 & 8 \end{pmatrix} + \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}. \checkmark$$

5.2 Expressing A^{-1} as a polynomial in A

Cayley–Hamilton turns the inverse into a polynomial — no augmented matrix required. Write

$$\chi_A(\lambda) = (-1)^n (\lambda^n + c_{n-1} \lambda^{n-1} + \dots + c_1 \lambda + c_0).$$

If A is invertible then $c_0 = (-1)^n \det A \neq 0$. Substituting A and isolating, then multiplying by A^{-1} ,

$$A^{-1} = -\frac{1}{c_0} (A^{n-1} + c_{n-1} A^{n-2} + \dots + c_1 I).$$

Worked example — inverse via Cayley–Hamilton. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $\chi_A(\lambda) = \lambda^2 - 4\lambda + 3$, so $A^2 - 4A + 3I = 0$, giving $3I = 4A - A^2 = A(4I - A)$. Hence

$$A^{-1} = \frac{1}{3}(4I - A) = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

5.3 Powers of a non-diagonalizable matrix

The same identity computes high powers without diagonalization. Cayley–Hamilton says A^n equals a combination of $I, A, \dots, A^{n-1}$; therefore **every** power A^k with $k \geq n$ reduces, step by step, to such a combination. This works whether or not A is diagonalizable — in particular for the shear and other defective matrices.

Worked example — reducing a high power. The shear $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has $\chi_A(\lambda) = (\lambda - 1)^2 = \lambda^2 - 2\lambda + 1$, so $A^2 = 2A - I$. Then $A^3 = A \cdot A^2 = 2A^2 - A = 2(2A - I) - A = 3A - 2I$, and inductively $A^k = kA - (k-1)I = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$ — the closed form for a power of a non-diagonalizable matrix, obtained with no eigenbasis.

6 Matrix inverse

We collect the two routes to A^{-1} , one computational and one algebraic. The first generalizes the Gaussian elimination of MAT11-3; the second is the Cayley–Hamilton method just seen.

6.1 Via Gaussian elimination

Augment A with the identity and row-reduce the **left** block to I ; the **right** block then holds A^{-1} :

$$(A \mid I) \sim \dots \sim (I \mid A^{-1}).$$

The justification is that each row operation is left-multiplication by an elementary matrix; the sequence that turns A into I multiplies to A^{-1} , and applying that same sequence to I produces A^{-1} . If the left block cannot be driven to I — a zero pivot appears — then A is **not invertible** ($\det A = 0$).

Worked example — inverse by elimination. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$:

$$\begin{aligned} \left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array} \right) &\sim \left(\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{array} \right) \sim \left(\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 0 & -3 & 1 & -2 \end{array} \right) \\ &\sim \left(\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 0 & 1 & -\frac{1}{3} & \frac{2}{3} \end{array} \right) \sim \left(\begin{array}{cc|cc} 1 & 0 & \frac{2}{3} & -\frac{1}{3} \\ 0 & 1 & -\frac{1}{3} & \frac{2}{3} \end{array} \right), \end{aligned}$$

so $A^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ — agreeing with the Cayley–Hamilton computation above.

6.2 Via Cayley–Hamilton

When $\det A \neq 0$, the formula $A^{-1} = -\frac{1}{c_0}(A^{n-1} + \dots + c_1 I)$ from the previous chapter gives the inverse directly. For small n or when you already have χ_A in hand (e.g. from an eigenvalue computation), this is often quicker than elimination, and it expresses A^{-1} in the conceptually useful form “a polynomial in A ”.

Exercises

13. Invert $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ by Gaussian elimination, then again via Cayley–Hamilton, and check the answers agree.
14. Use Cayley–Hamilton to write A^{-1} for $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.
15. For which values of t is $\begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}$ invertible? Find the inverse when it exists.

7 Applications

The machinery now pays for itself. Two application areas in the OP — graphs and Markov chains — are both, at heart, the study of A^k as k grows, exactly what diagonalization computes.

7.1 Graph adjacency matrices and walks

Definition 17 The **adjacency matrix** A of a graph on vertices $1, \dots, n$ has entry a_{ij} equal to the number of edges from vertex i to vertex j .

Proposition 18 For every integer $k \geq 1$, the entry $(A^k)_{ij}$ is the number of **walks of length k** (sequences of k consecutive edges) from vertex i to vertex j .

Proof. By induction. For $k = 1$ this is the definition. Assuming it for k , a walk of length $k + 1$ from i to j is a walk of length k from i to some intermediate m , followed by one edge $m \rightarrow j$. Summing over m , $(A^{k+1})_{ij} = \sum_m (A^k)_{im} a_{mj}$, which is exactly the matrix product — the rule of matrix multiplication is the rule for concatenating walks. $\square$

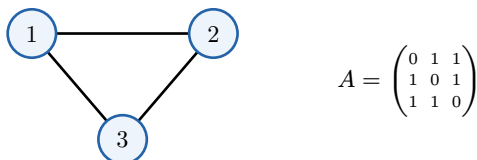


Figure 5: A triangle graph and its adjacency matrix. There is no edge from a vertex to itself, so the diagonal is 0. The number of walks of length k between any two vertices is read off A^k ; e.g. $(A^2)_{11} = 2$ counts the two length-2 walks $1 \rightarrow 2 \rightarrow 1$ and $1 \rightarrow 3 \rightarrow 1$.

The asymptotics of walk counts are governed by the **dominant eigenvalue** of A (the one largest in absolute value): as k grows, A^k is dominated by the $\lambda_{\max}^k$ term in its diagonalization, and the corresponding eigenvector captures the proportion in which the walk-count concentrates across vertices.

7.2 Markov chains via stochastic matrices

A Markov chain models a system hopping between n states, where the next state depends only on the current one. Its transitions are encoded by a stochastic matrix, and its long-run behaviour is — once again — an eigenvalue problem.

Definition 19 A square matrix P with non-negative entries is **column-stochastic** if every column sums to 1 (and **row-stochastic** if every row sums to 1). A **probability distribution** is a vector with non-negative entries summing to 1.

We use the column-stochastic convention: p_{ij} is the probability of moving **to** state i **from** state j , and a distribution x_k evolves by $x_{k+1} = Px_k$.

Proposition 20 Every column-stochastic matrix P has $\lambda = 1$ as an eigenvalue. A **stationary distribution** is a probability distribution π fixed by P ,

$$P\pi = \pi,$$

i.e. an eigenvector for the eigenvalue 1.

Proof. Each column of P sums to 1, which says $\mathbf{1}^T P = \mathbf{1}^T$ where $\mathbf{1} = (1, \dots, 1)^T$. So $\mathbf{1}$ is an eigenvector of P^T for the eigenvalue 1. Since $\det(P - I) = \det(P^T - I)$ (transpose invariance), 1 is an eigenvalue of P as well, guaranteeing a fixed distribution π . $\square$

Worked example — a two-state chain converging to equilibrium. Two cities; each year a fraction of residents migrate. Let

$$P = \begin{pmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{pmatrix},$$

so 80% of city-1 residents stay and 30% of city-2 residents move to city 1, etc. (columns sum to 1). The eigenvalues are 1 and 0.5 ($\text{tr } P = 1.5$, $\det P = 0.5$). The eigenvalue 1 gives the stationary distribution; solving $(P - I)\pi = 0$ and normalizing,

$$\pi = \begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix}.$$

The second eigenvalue 0.5 has $|0.5| < 1$, so its contribution decays.

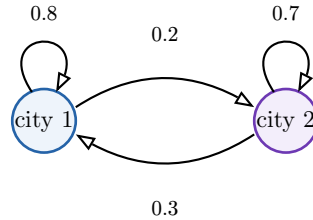


Figure 6: The two-state Markov chain. Arrows carry transition probabilities; each city's outgoing probabilities (its column of P) sum to 1. From any starting split, the population converges to the stationary distribution $\pi = (0.6, 0.4)$ — the eigenvector for eigenvalue 1.

Theorem 21 (Convergence to stationarity) Suppose P is column-stochastic and diagonalizable, with 1 as a **dominant** eigenvalue — every other eigenvalue μ satisfying $|\mu| < 1$. Then for **any** initial distribution x_0 ,

$$x_k = P^k x_0 \rightarrow \pi \quad \text{as } k \rightarrow \infty,$$

where π is the stationary distribution.

Proof. Write $P = QDQ^{-1}$ with eigenvalues $1, \mu_2, \dots, \mu_n$ on the diagonal. Then $x_k = P^k x_0 = QD^k Q^{-1} x_0$, and $D^k = \text{diag}(1, \mu_2^k, \dots, \mu_n^k)$. Since $|\mu_i| < 1$ for $i \geq 2$, every $\mu_i^k \rightarrow 0$, so D^k tends to $\text{diag}(1, 0, \dots, 0)$. Hence x_k tends to the component of x_0 along the eigenvector for 1 — which, normalized, is π , independent of x_0 . $\square$

This is the same principle as Fibonacci's φ and the graph's dominant eigenvalue: **the largest eigenvalue dictates the long run**, and diagonalization is what lets us see it.

Worked example — watching the chain settle. Start everyone in city 2: $x_0 = (0, 1)^T$. Then $x_1 = Px_0 = (0.3, 0.7)^T$, $x_2 = (0.45, 0.55)^T$, $x_3 = (0.525, 0.475)^T$, ..., marching steadily toward $\pi = (0.6, 0.4)^T$. Starting instead from $x_0 = (1, 0)^T$ leads to the **same** limit — the chain forgets where it began.

Exercises

16. Write the adjacency matrix of the path graph $1 - 2 - 3$ and compute the number of length-2 walks from 1 to 3.
17. Verify directly that $\pi = (0.6, 0.4)^T$ satisfies $P\pi = \pi$ for the migration matrix above.
18. Find the stationary distribution of $P = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & \frac{3}{4} \end{pmatrix}$.
19. A stochastic matrix has eigenvalues 1 and -1 . Does $P^k x_0$ converge for every x_0 ? Explain using the convergence theorem's hypotheses.

8 Similar matrices and change of basis

We close with the conceptual capstone that justifies the whole course: the same linear transformation can wear many matrices, and diagonalization is the search for its simplest one.

8.1 Similarity

Definition 22 Two matrices $A, B \in M_n(\mathbb{R})$ are **similar** if there is an invertible P with

$$B = P^{-1}AP.$$

Proposition 23 (Similarity invariants) Similar matrices have the same characteristic polynomial, and hence the same eigenvalues (with multiplicities), the same determinant, and the same trace.

Proof. Using multiplicativity and $\det(P^{-1}) = 1/\det(P)$,

$$\det(B - \lambda I) = \det(P^{-1}(A - \lambda I)P) = \det(P^{-1}) \det(A - \lambda I) \det(P) = \det(A - \lambda I).$$

Equal characteristic polynomials give equal eigenvalues; trace and determinant are read off that polynomial. $\square$

These shared quantities are exactly the data that does **not** depend on the choice of coordinates — the intrinsic fingerprint of the transformation, promised in the preface.

8.2 Why similarity means “same map, new basis”

Recall from MAT11-3 that a single linear map f has different matrices in different bases, and that a change of basis is recorded by an invertible matrix P (its columns the new basis vectors in old coordinates).

Proposition 24 If A is the matrix of a linear map f in one basis and P is the change-of-basis matrix to a second basis, then the matrix of the **same** map f in the second basis is $P^{-1}AP$.

So **similar matrices are precisely the matrices of one transformation in different coordinate systems**. The eigenvalues, trace, and determinant are invariant because they belong to the transformation, not to the coordinates.

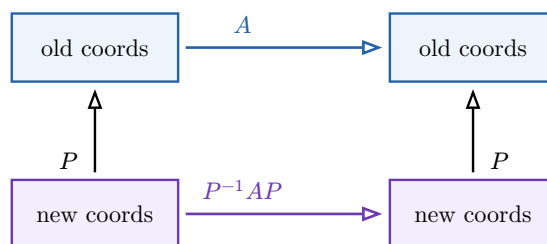


Figure 7: One map, two coordinate systems. Going across the top applies the map in old coordinates (A); going down–across–up translates to new coordinates, applies the map, and translates back ($P^{-1}AP$). Both routes describe the **same** transformation, so A and $P^{-1}AP$ are similar.

8.3 Diagonalization as the best basis

In this light, diagonalization is the climax of the entire two-course sequence. To say $A = PDP^{-1}$ — i.e. A is **similar to a diagonal matrix** — is to say that **in the basis of eigenvectors** the transformation is nothing more than an independent scaling of each axis by its eigenvalue. The arbitrary numbers of A were an artefact of bad coordinates; the eigenvalues in D are the transformation’s true content.

The arc in one breath. The **determinant** detects invertibility and measures signed volume. Setting $\det(A - \lambda I) = 0$ finds the **eigenvalues**; their eigenspaces are the directions A only scales. When the **geometric multiplicities** fill out the **algebraic** ones, a basis of eigenvectors exists and $A = PDP^{-1}$ is **diagonalizable**, making $A^k = PD^kP^{-1}$ — and hence recurrences, walk counts, and Markov limits — easy. **Cayley–Hamilton** supplies inverses and powers even when diagonalization fails. And the closing truth: **similar** matrices are one transformation in different bases, so diagonalization is simply the act of choosing the coordinates in which a map is as simple as it can be.